VIKRANT KUMAR

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PROFESSIONAL SUMMARY

M.Tech student in Data Science Engineering with hands-on research experience in deep learning, computer vision, and AI model development. Currently contributing to an ANRF-funded government research project on cancer detection using CNN architectures, transfer learning, and medical image analysis, with practical experience in model optimization and fine-tuning techniques relevant to modern LLM adaptation. Actively expanding expertise in Large Language Models (LLMs), parameter-efficient fine-tuning, and knowledge graphs. Passionate about leveraging AI and data-driven technologies to build innovative products, solve real-world business challenges, and contribute to impactful research and software development.

EDUCATION

M.Tech – Data Science & Engineering — **CGPA: 8.32**

National Institute of Technology, Silchar — 2025–2027 (Ongoing)

B.Tech – Computer Science & Engineering — **CGPA: 7.04**

Assam University, Silchar — 2021–2025

RESEARCH & INTERNSHIP EXPERIENCE

Research Intern – Cancer Detection (ANRF Ministry Project)

NIT Silchar — May 2026 – Present

- Working on a government-funded ANRF project for cancer detection using deep learning on the BUSI (Breast Ultrasound Images) dataset — involving CNN and transfer learning model development.
- Designed and evaluated multi-model architectures (VGG16, DenseNet121, MoE frameworks); applied data augmentation, focal loss, and class-balancing strategies to handle real-world data skew.
- Explored fine-tuning strategies for pretrained models — directly transferable to LLM domain adaptation.
- Delivered a technical session on CNNs during a five-day ANRF-organised biomedical deep learning workshop at NIT Silchar (June 2026); received certificates for both participation and contribution.
- Authored research documentation covering experimental design, model evaluation (accuracy, F1, Grad-CAM), and performance analysis — building scientific writing skills applicable to AI research publishing.

Web Development Intern

Main Flow — May 2024 – Jul 2024

- Built dashboards using JavaScript and REST APIs; gathered stakeholder requirements and tracked deliverables.

Web Development Intern

Internshala — Jul 2023 – Oct 2023

- Developed responsive web apps (HTML, CSS, JavaScript, PHP, MySQL); implemented authentication and optimised SQL queries.

PROJECTS

Breast Ultrasound Image Classification (Ongoing Research) — Python, TensorFlow, PyTorch, Keras

- Architected a Mixture-of-Experts (MoE) framework for three-class BUSI ultrasound classification, combining specialised CNN experts with a trainable gating network.
- Addressed class imbalance with focal loss and class-aware weighting.
- Accelerated training and experimentation on an NVIDIA RTX A4000 using CUDA, automatic mixed precision, and gradient scaling.
- Applied modular fine-tuning, selective parameter optimisation, and metric-driven evaluation—foundational practices for parameter-efficient LLM adaptation, including LoRA, and alignment-oriented training workflows.

Lung Cancer Classification Using Transfer-Learning — Python, TensorFlow, Keras, OpenCV [GitHub]

- Fine-tuned and benchmarked five pre-trained CNN backbones—ResNet50, EfficientNetB3, InceptionV3, DenseNet121, and VGG16—for multi-class lung CT image classification.
- temImplemented image preprocessing, augmentation, selective layer unfreezing, and K-Fold cross-validation to improve generalisation on limited medical-image data.

- Built a metric-driven evaluation pipeline using accuracy, precision, recall, F1-score, and Grad-CAM visualisation; applied selective parameter optimisation practices transferable to parameter-efficient LLM fine-tuning.

Training & Placement Cell Website — React, Vercel [tpc-cse-aus.vercel.app]

- Built and deployed a student portal for job/internship listings; managed requirements and deployment end-to-end.

TECHNICAL SKILLS

AI / ML & Deep Learning	CNNs, Transfer Learning, MoE Architectures, Computer Vision, Model Evaluation, TensorFlow, PyTorch, Keras, Scikit-learn
Gen AI (Active Study)	Transformer Architecture, LLM Prompting, RLHF, LoRA Fine-Tuning, Knowledge Graphs, Text Generation Quality
Data Analysis	Pandas, NumPy, SQL, EDA, Matplotlib, Seaborn, Power BI
Programming Languages	Python (primary), C, C++, Java
Core CS	Data Structures & Algorithms, OOP, Operating Systems, DBMS
Tools & Platforms	Git/GitHub, VS Code, Jupyter Notebook, Google Colab, MS Excel, Docker (basic)
Web Technologies	HTML, CSS, JavaScript, React, REST APIs
Methodologies	Agile/Scrum, Research Documentation, Experimental Design, Scientific Writing

CERTIFICATES

- Volunteer and Participant Certificate — ANRF-Sponsored Short-Term Workshop on *Biomedical Applications Using Deep Learning Techniques*, Department of CSE, NIT Silchar, June 2026; delivered a technical session on CNNs and BUSI breast-ultrasound image classification. [Certificate]

LEADERSHIP & ACTIVITIES

- Assisted the placement coordination team at NIT Silchar by updating placement records and sharing recruitment-related information with company recruiters via email.
- Supported workshop operations at NIT Silchar, including participant registration, certificate generation, and technical-session logistics.
- Active participant in cricket and badminton.